On Job Training-02

1.Select any one job role of his/her interest in an organization for next 4 weeks and submit a report to the training supervisor and copy to the cohort owner focusing on:

Job Role: AIML and Software Development Intern
Responsibility: Testing and Optimization
Phase: Enhancement

The Genetic Disease Prediction System is a machine learning–based web application designed to assist users in analysing the probability of genetic disease risk based on personal and genetic indicators. The system integrates frontend user interfaces, backend processing using Flask, and machine learning models to provide predictive insights in a structured and user-friendly manner.

The application allows users to create accounts, securely log in, upload datasets, and perform predictive analysis using trained machine learning models. The system focuses on improving accessibility by enabling both dataset upload and sample dataset usage, ensuring usability even for users without prior data.

The project combines concepts from web development, data pre-processing, and machine learning to create a unified system capable of handling real-world data scenarios. It also emphasizes user experience through enhanced interface design, validation mechanisms, and visual result representation.

i) Application of Skills and Technical Knowledge

- ✓ **Python & Backend Development**

Python was extensively used as the core programming language for backend development. The Flask framework was utilized to create REST APIs for handling user authentication, dataset processing, and machine learning prediction requests. Backend routes were structured to manage operations such as login validation, dataset upload, pre-processing, and prediction execution. Modules such as request, jsonify, and routing mechanisms were used to enable communication between frontend and backend systems.

- ✓ **Database Integration (Firebase)**

Firebase was used as a cloud-based database solution to manage user authentication and store personal user details. Real-time database services were utilized to store and retrieve user information dynamically.

Authentication mechanisms ensured that user credentials were securely validated.

CRUD operations such as account creation, update, and deletion were implemented, allowing seamless interaction between the application and the database.

- ✓ **Frontend Development**

The user interface was carefully designed using modern UI principles such as:

- Card-based layouts
- Two-column responsive structure
- Consistent colour themes (navy blue, teal, white)
- Interactive elements with animations

- ✓ **Machine Learning Implementation**

Machine learning models were implemented using libraries such as Scikit-learn and XGBoost. The system initially used Logistic Regression but was later enhanced by evaluating multiple models including:

- Support Vector Machine
- Random Forest
- Gradient Boosting
- XGBoost
- The final model selected was XGBoost due to its ability to handle non-linear relationships and provide more dynamic probability outputs.

- ✓ **Data Pre-Processing Techniques**

Data pre-processing played a crucial role in ensuring model accuracy. The following steps were implemented:

- Handling missing values
- Removing duplicate records
- Data normalization and scaling
- Encoding categorical variables
- Feature selection and target separation

ii) Performance on Assigned Tasks and Projects

During the internship period, multiple modules of the system were successfully developed and integrated.

- A complete user authentication system was implemented with login, signup, CAPTCHA validation, and error handling mechanisms.

- Dynamic dashboard functionality was developed to display user-specific data and system operations.
- Dataset upload and pre-processing modules were created, allowing users to upload and process real-world datasets.
- Machine learning integration was successfully implemented, enabling prediction generation based on user input.
- Visualization components such as risk gauge meters and charts were developed to present prediction results clearly

Additionally, several debugging and optimization tasks were performed, including resolving layout issues, improving validation mechanisms, and enhancing backend logic.

iii) Value Addition to the Organization

The project contributed significant value by developing a complete end-to-end system for genetic disease prediction.

One of the key contributions was improving system accessibility by introducing a built-in dataset download feature. This eliminated the dependency on external datasets and allowed users to immediately interact with the system.

➢ **User experience was enhanced through:**

- Improved UI design
- Clear error messages
- Keyboard shortcuts
- Automated form validation

The integration of machine learning models into a real-time application demonstrated the practical applicability of theoretical knowledge.

Additionally, the system introduced educational components that guide users in understanding genetic data, thereby increasing the usability and interpretability of the application.

iv) Job Role Understanding and Relevance

During the internship, the role involved working as an AI, Machine Learning, and Software Development Intern. Responsibilities included developing backend services, designing user interfaces, implementing machine learning models, and integrating all components into a functional application.

The role required understanding both theoretical and practical aspects of system development, including data handling, model training, and user interaction design.

Use Case-02

2. Document a Use Case on a Task where he/she is working as an intern

Use Case Documentation

2.1 Use Case Title

Genetic Disease Risk Prediction System

2.2 Actors

- User (Doctor / Researcher / Admin)
- System

2.3 Inputs

✔ Age
✔ Gender values
✔ Family History
✔ SNP values
✔ Gene expression

2.4 Process

1. System Execution

The backend and frontend are initialized in command prompt to run the system locally

2. Authentication

If you don't have an account, click on create account to sign up with the system with your correct details and Log in using your unique Userid and password. Both these pages have been enhanced by using correct validation and correct error messages

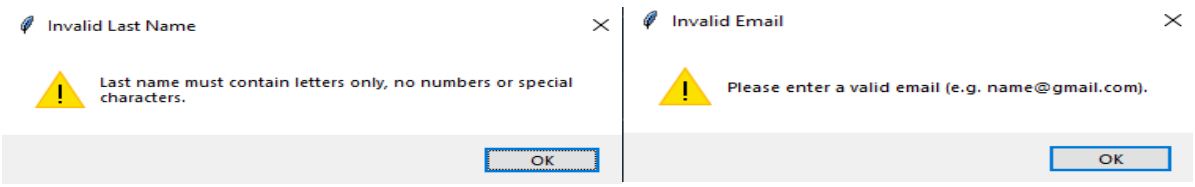


Figure 3.2.1: Error authentication messages

Previously, the system was unable to process spaces in names, and unable to verify the correct format for all the input fields.

3. Dashboard

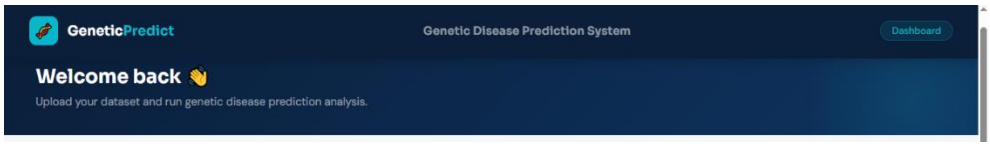


Figure 3.2.2: Enhanced Dashboard page

Dashboard page is now enhanced with hero cards, teal headers, logo icon, name indicating the system usage and information of page on the right.

4. User Management

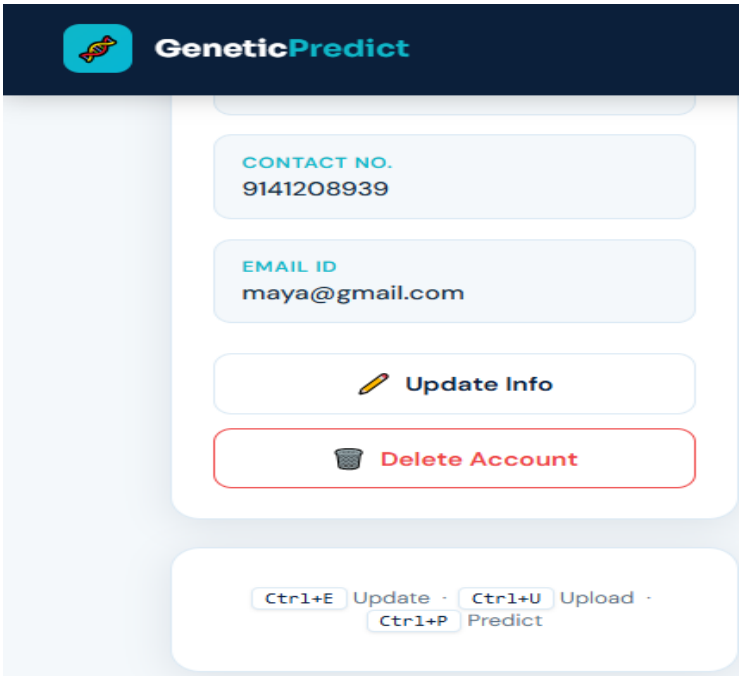


Figure 3.2.3.1: Added delete account button

Personal information now has a delete account button to delete your personal information and account permanently, it prompts for a password for authentication and comes out of the dashboard page.

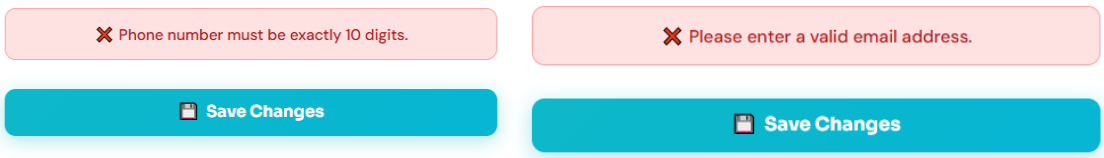


Figure 3.2.3.2: Update page with authentication

If you type in the wrong format for names, email or phone number, and error message will be displayed and your details won't be modified now

5. Dataset Module

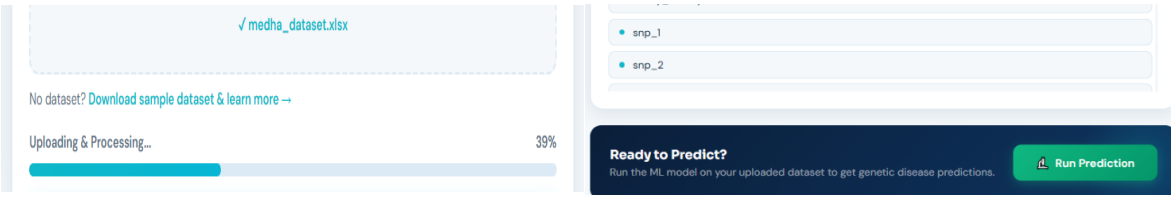


Figure 3.2.4: Enhanced dataset upload

Enhanced scrollable properties for features and a new link for people who don't have their own dataset.

6. Dataset Information Page

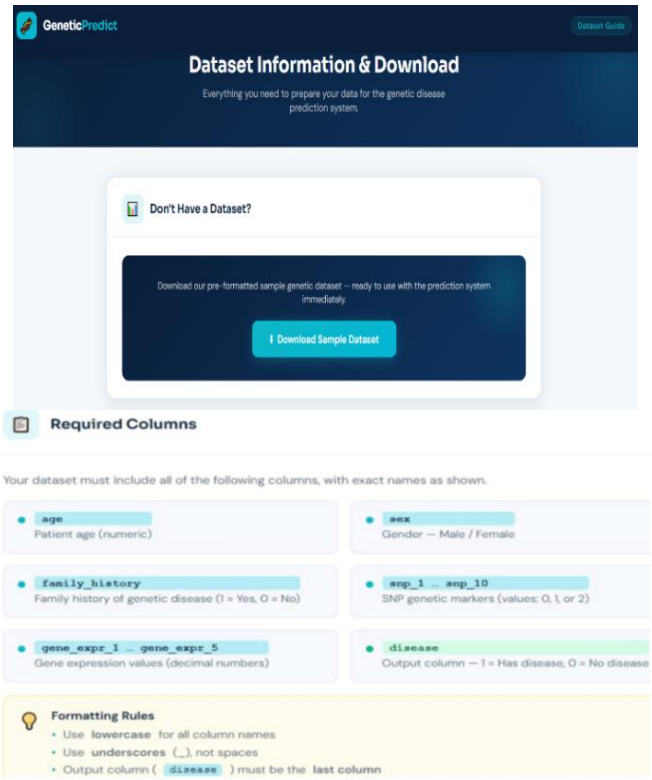


Figure 3.2.5: Dataset Information page regarding format and columns

7. ML Prediction Module

Prediction page with a “Fill Sample Data” button for first time users and genetic information page for people who don’t know their SNP values. Once you fill all the input fields you can run the prediction button to find your result.

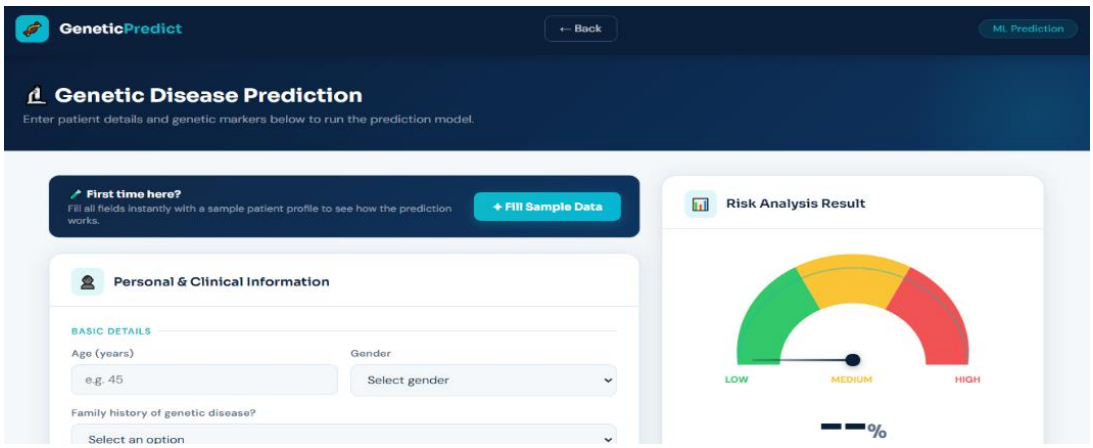


Figure 3.2.6.1: Prediction page with sample input

Your input must be valid, if not, the system will not move forward and will show the error in a red border around the box, if valid, the input fields will be in green border.

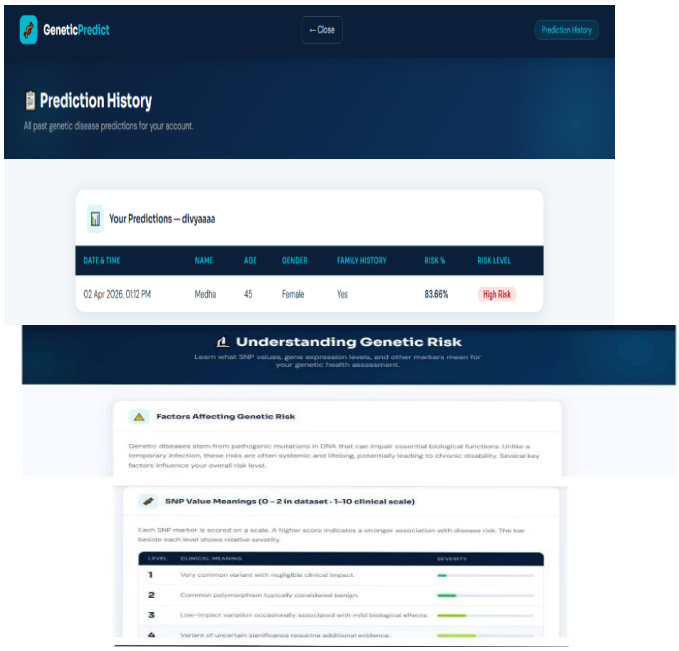


Figure 3.2.6.2: Genetic Information and View History Page

For people who don’t understand the model or do not know their SNP values, a new link has been created which will take them to a new page consisting of gene data. Clicking the View History button will take them to a new page consisting of their previous predictions.

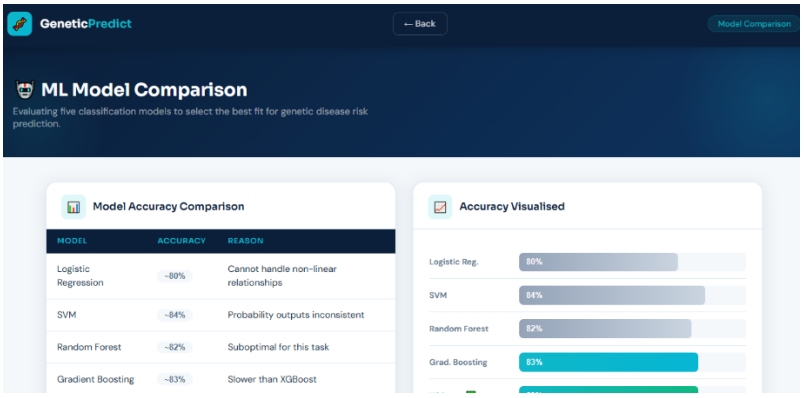


Figure 3.2.6.3: Model Comparison Page

For people who want to view their model accuracy and selection, a new link called View Model Comparison is created clicking on which will take them to a new page consisting of model accuracy, comparison and classification report.

2.5 <u>Output</u>

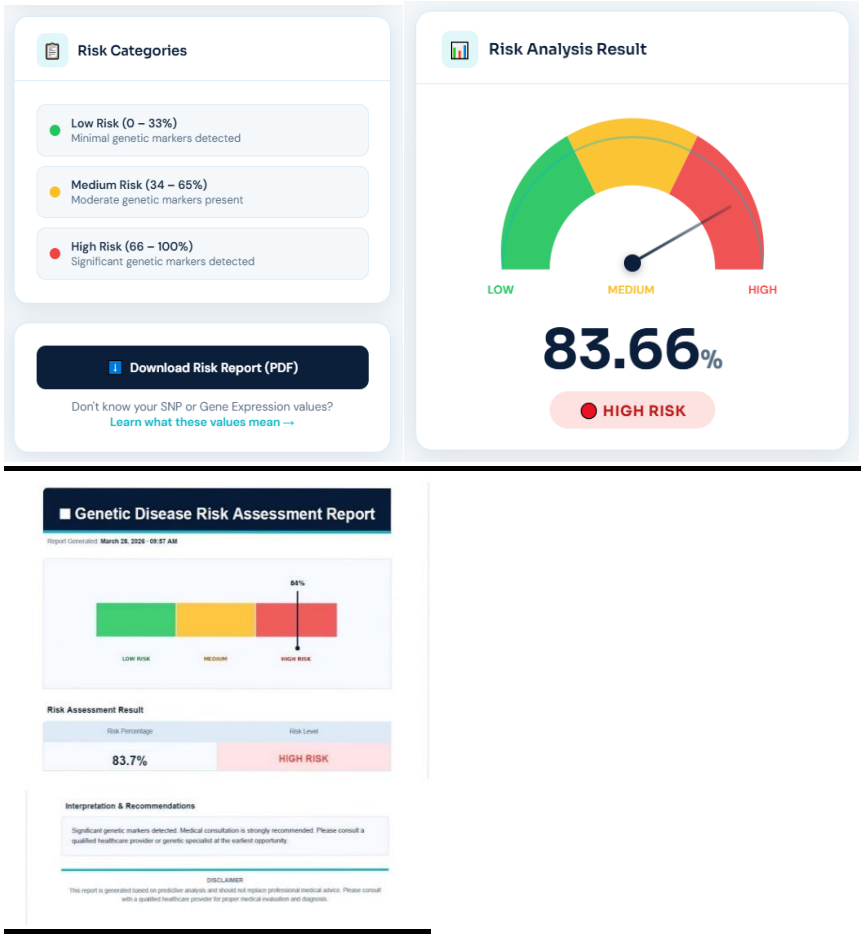


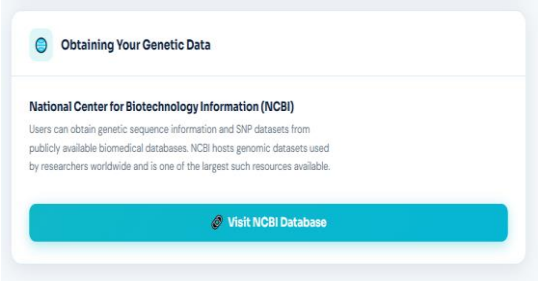
Figure 3.2.7: Final Output – Risk prediction (low, medium, high) with a report + chart

Explanation of Risk categories, Risk visualization and a Risk report for people to store it in offline.

2.6 External Integration

The system integrates external knowledge sources such as the National Center for Biotechnology Information to help users understand genetic data and obtain biological datasets and SNP values

Figure 3.2.8: NCBI



2.7 Testing

A comprehensive system testing and validation process was carried out to ensure the reliability, accuracy, and robustness of the developed Genetic Disease Prediction System.

The testing process began with the authentication module. The login functionality was validated using multiple scenarios including invalid user ID, incorrect password, and incorrect CAPTCHA inputs. Similarly, the signup module was tested to verify successful account creation and proper handling of duplicate user IDs.

Profile management features such as updating personal details and account deletion were tested and the dataset module was validated by testing dataset upload functionality. The dataset information page was also verified for proper download of the sample dataset and correctness of formatting instructions.

The prediction module was tested to confirm that the machine learning model generated outputs correctly based on user inputs. The visualization components, were tested for accuracy and responsiveness. The genetic information page and prediction history module were verified to ensure proper navigation and correct retrieval of stored user-specific prediction records.

Model performance was evaluated using standard metrics such as accuracy, precision, recall, and classification report. This ensured that the final selected model provided consistent and reliable prediction results.

Overall, the testing process confirmed that all modules were functioning correctly and the system was stable for deployment.

Example 1:

Test Case 01 –Sign Up

- Input: Valid details
- Expected: Account created
- Result: Success message

Status: **PASS**

2.8 Deployment

The final phase of the project involved deployment of the application to make it accessible online. Prior to deployment, all project files were carefully reviewed, organized, and consolidated to ensure proper structure and maintainability.

Version control was implemented using Git, where the project repository was initialized, staged, committed, and pushed to a remote repository on GitHub.

Deployment of the system on cloud platforms like Render and Railway was explored, however full scale deployment was not implemented due to platform constraints and resource limitations.

2.9 Advantages

Figure 3.2.9: Key Advantages of the System

2.10 Limitations

Despite its functionality, the system has certain limitations that need to be considered.

Figure 3.2.10: Key Limitations of the System

2.11 Future Enhancements

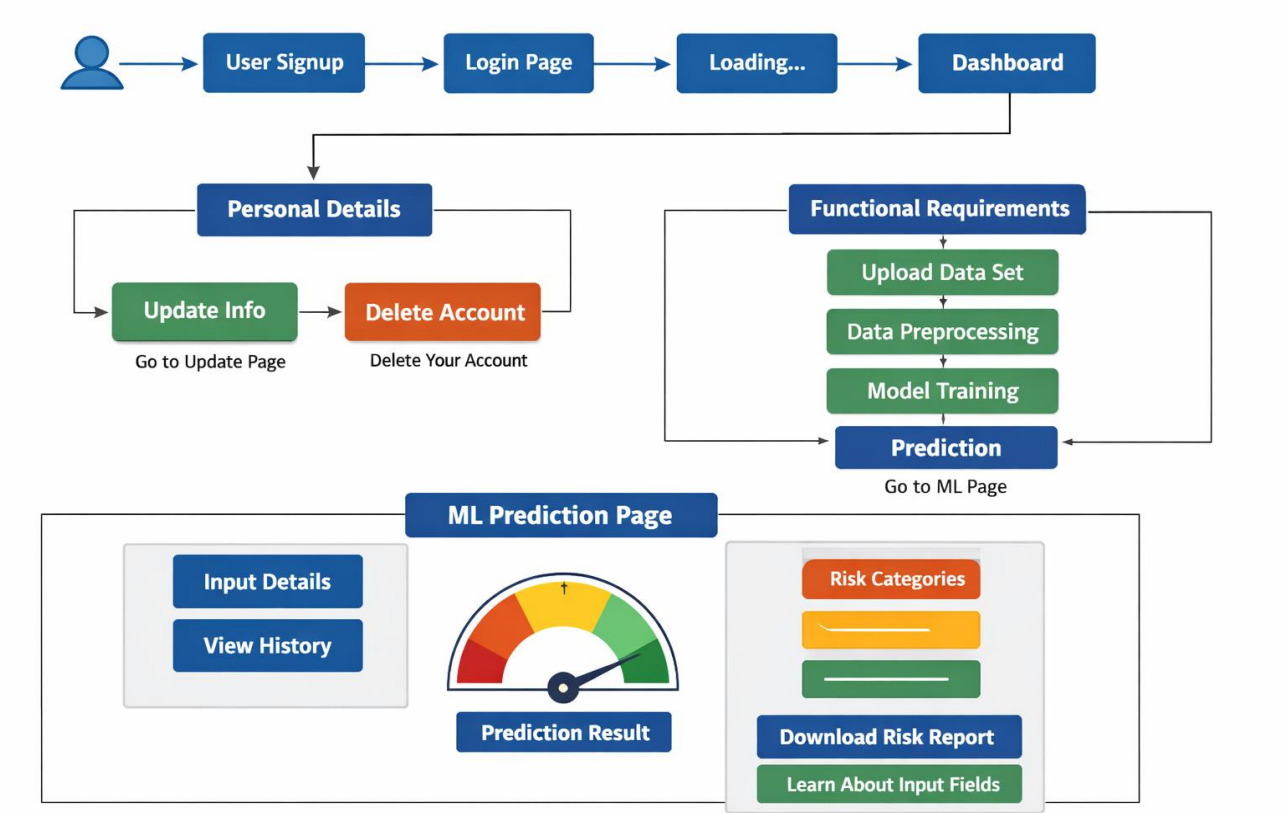


Figure 3.2.11: Current System Flow

The system can be further improved by implementing several enhancements in future development phases.

One major improvement would be the integration of larger and more diverse real-world genetic datasets with automatic dataset balancing. This would improve model accuracy and allow the system to generalize better across different populations.

The application can also be enhanced by incorporating additional machine learning and deep learning models such as neural networks to capture more complex biological relationships.

The user interface can be further improved by adding multi-disease prediction support and more interactive visualizations and personalized insights based on user data.

Integration with healthcare platforms or wearable devices could also allow real-time data collection and continuous monitoring.

Cloud based scalable deployment for larger datasets and addition of chatbot support for guiding users through the system.

Security features such as encrypted authentication, secure APIs, and role-based access control can be implemented to ensure data privacy and protection.

Finally, multilingual support and mobile application development can be introduced to increase accessibility for a wider range of users.

Conclusion

The Genetic Disease Prediction System successfully demonstrates the integration of web development and machine learning techniques to create a functional and user-friendly application.

Throughout the development process, various components such as user authentication, dataset handling, data pre-processing, model training, and result visualization were implemented and integrated into a cohesive system.

The project highlights the practical application of machine learning in the healthcare domain, particularly in predicting genetic disease risks based on input parameters.

In addition to technical implementation, the system also emphasizes usability and accessibility through enhanced user interface design, validation mechanisms, and educational support features.

Although certain limitations exist, the project provides a strong foundation for further development and improvement. Overall, the system serves as an effective demonstration of how modern technologies can be combined to solve real-world problems and deliver meaningful insights.